

“How much connection did you get?” Drinking Journaling Data through Hot Cocoa

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Figure 1: KuKou box on the left, connected journaling device on the right.

ABSTRACT

What if you could get a taste of your own thoughts? We propose a new system in HCI that uses computational tools to map text onto flavors and thus allow users to “eat their words” in the form of a custom hot cocoa blend created from written journal entries, helping people engage in reflection. Our more open-ended approach to computer-augmented reflection explores new tangible mediums and methods to facilitate a sense of playful fluidity in machine interactions. A deep learning pipeline is developed that finds meaning within user journals by identifying similar poems/stories/texts in a pre-collected corpus. A dispenser outputs a drink mix with flavors

based on these discovered meanings, and prints a receipt to help users make sense of their experience. To encourage disclosure and create a reflective mood, we model our experience based on journaling at an intimate cafe. Our preliminary evaluation demonstrates strong promise in this approach.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design theory, concepts and paradigms.**

KEYWORDS

edible data, machine learning, natural language processing, reflection, taste interaction, interactive installation, personal informatics, tangible user interfaces

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1 INTRODUCTION

Ubiquitous computing tools have given rise to the field of personal informatics, in which technology has been developed to capture personal data and help people reflect on their lives [12]. The goal of much personal informatics work allows users to record behaviors through data collection (for instance, data on exercise habits), reflect on the data, and then drive action [2]. Reflection is known to be key to this process; rather than rotely revisit collected data, such systems aim to synthesize and transform this data in a way that allows users to draw more personal insights and thus lead to some sort of new understanding or behavior [13].

Existing systems may not fully capture user needs in this reflection stage. Current designs of such systems present their data in a highly "objective" and numerical way; there is little space for ambiguity and subjective interpretation. However, users desire such subjectivity and open-endedness. People identify open-ended processes such as re-interpretation and introspection to be a necessary component of this reflective process [11] [4]. In fact, the way that personal informatics systems show emotional data may actually alienate and turn away users due to overly quantified and "objective" representations [1] [9].

Thus, for personal informatics systems to better facilitate reflection, they may try to encourage key processes like introspection and emotional understanding. To avoid the pitfalls described above, we require a better way of summarizing and presenting information. By leveraging ambiguity and subjectivity as a resource in the presentation of personal data, we hope to encourage continued engagement with the system [7] and facilitate deeper reflection.

With this as our goal, we take a critical approach to data representation in personal informatics by exploring more subjective representations of qualitative personal information - specifically, hot chocolate representations of journal entries.

Why journals, and why hot chocolate? We choose to represent journal entries rather than something such as health data due to its uniquely qualitative and emotional nature. The emotions and experiences reflected in personal journals are uniquely difficult for personal informatics systems to capture and represent for the reasons described earlier. Furthermore, we believed that edible representations matched well thematically with journal text. Journals tie together emotions and memories; flavor is able to do the same. In almost every language, there are rich verbal metaphors tying flavor to experience (we savor our memories, say bittersweet goodbyes, have arguments gone sour, enjoy the warm spices of the holiday season, etc). This natural connection of language, flavor, and memory made edible representations a compelling representational medium for us to explore. We settled on hot chocolate mix due primarily to practical concerns described in the design section.

This line of thinking inspired the creation of the KuKou box, a system that dispenses a custom hot chocolate drink based on the text of compiled journal entries. It contains three major components: a journal interpretation tool to assign "meaning" to collected journal entries, a drink dispenser that dispenses a custom hot chocolate mix based on the extracted journal interpretations, and a receipt that helps the users make sense of their experience. We do a preliminary evaluation and identify future directions for this work.

2 BACKGROUND AND RELATED WORK

2.1 Limits of existing personal informatics systems for reflection

Prior research in personal informatics has isolated five components: preparation, collection, integration, reflection, and action [13]. We are interested in the reflection stage, defined as the stage in which different strategies are explored for synthesizing, presenting, and transforming data [13]. At this stage, users gain some sort of new insight from collected information. This typically involves some form of transformation or deeper inner analysis rather than a rote reproduction of the data. However, existing applications of personal informatics may struggle to actively facilitate the open-ended processes that people value [11] [4]. This reflects a broader values-level problem in which computing may have difficulty engaging with the ambiguous and "messy" components of human life [3] due to an emphasis on formal systems and inflexible categorizations.

This is particularly relevant when it comes to reflection that engages more emotional content, such as journal entries. When computers are involved in assigning emotions to content or presenting information about emotions, users find that computer interpretation and presentation of emotions is alienating, opaque, too objective, and limiting [1] [9]. For instance, conventional sentiment analysis tools like text2emotion establish a small set of "basic emotions" (happy, anger, sad, fear, surprise) and can not be extended beyond that. Such models are also opaque to the average user, as the final output of such tools is often just an unexplained collection of numbers.

2.2 The potential of edible representations and journaling data

Edible interfaces are a source of active research, particularly for representing data [21]. Often, they are used to represent quantifiable data such as heart rate [10] [6], laughter frequency [20], or personal information [18] through physical qualities such as size, volume, shape, and design. In contrast, leveraging flavor for semantic meaning is more difficult because flavor is uniquely semantically imprecise and open to interpretation. Associations of flavor/texture to meaning are highly subjective from person to person [14]; for instance, the flavor of cinnamon may call back memories of a warming cider in autumn for some, while invoking a refreshing horchata in the summer for others. Moreover, an individual's perception of the same flavor can vary drastically in intensity or quality from day to day [16].

However, we perceive this subjectivity and uniquely emotional quality to be a strength when it comes to the process of open-ended reflection. Flavor is strongly tied to memory and experience; the experience of a particular smell or flavor can very powerfully drive reminiscence, and thus reflection, on associated experiences [8] [15]. Since interpretations of flavor are so unique from person to person, it can induce highly personal associations. Food and flavor have also been used to facilitate reflections on empathy for social issues [5] and play [22] [17], once again demonstrating flavor's power to present information in an interactive way.

Moreover, journaling data presents a unique opportunity for edible data. As mentioned earlier, bio-sensory data such as heart rates

or step counts make up much of personal informatics research [10] [6]. Journal entries, in contrast, tend to represent more subjective mental and emotional states that people already naturally reflect on. This makes them a natural fit for our edible representations.

3 DESIGN PROCESS

We explore a critical perspective on personal informatics through edible information representations. The system diagram in Figure 2 shows the three major components (the journal interpretation, drink creation, and receipt design), as well as the interaction flow. Computation was done via a connected computer, and mechanical control was done via an Arduino Uno. These three components were mounted in a box that was designed to evoke a sense of ritual and mystique, calling forth the feeling of a tea or coffee shop, to encourage people to enter a reflective mindset.

3.1 Journal Interpretation

3.1.1 Summary. This module attempts to "summarize" emotions of journal text for presentation to the user. This would typically call for standard sentiment analysis tools. However, this was a method explicitly identified in the Background section as being alienating, too rigid, and opaque to the end-user.

We develop a new method which computationally matches journals to a corpus of human-labelled texts via a deep learning approach. In summary, given a written sentence in a journal, we would compare this sentence to every sentence in our corpus to find sentences that were semantically similar. Emotion would then be assigned to the journal text based on the emotions associated with the closest documents, and a user would receive both the scores and a "relevant passage" reflecting the document (or section of the document) closest in meaning to their own journal entry. Though this method might not have been as "accurate" as sentiment analysis, we believed it would have a few advantages. First, it would be somewhat contestable and interpretable - instead of receiving just a few scores back, the user could read the relevant passage, think about the connections between their writing and the passage, and decide whether this connection was spurious or not. Second, it would feel generally more delightful and "human"; rather than feel coldly judged by a machine, users could feel that their stories and experiences were related to those of other human beings.

3.1.2 Implementation details. The corpus was constructed and labelled manually by looking for poems/stories/internet comments that were about rest, stress, loneliness, and connection.

Sentences from the journal entries as well as the labelled corpus were converted to numerical vectors via Transformer neural networks. In such an approach, numerically similar vectors represent similar semantic meanings [19]; for instance, the sentence "The wedding festivities went on for hours" would have a similar numerical representation to the sentence "The party hasn't stopped all night".

Then, the vector representation of the journal entries were compared to the corpus to find the documents that were, on average, the shortest distance away from the journals. This reflected documents that were approximately similar in tone and meaning to the journals. We then took a weighted sum of the documents that were closest to the journal entry to assign the journal a "score" (e.g. if the

5 closest documents had 5 pertaining to rest and none pertaining to its opposite, stress, then we would give it a high rest score).

We also added an extra measure - "chewiness" - based on how many words the user disclosed. If they disclosed a lot, then we judged that they had "a lot to chew on".

The output contained three axes - a rest-stress axis, in which negative values referred to stress, a connection-loneliness axis, in which negative values referred to loneliness, and a chewiness axis, in which a higher number means "more to chew on". Additionally, the output contained the document that was judged to be the "closest" to the journal entry and thus most impactful to the assigned meanings; this provides the user some insight into how the scores were assigned / how the model conducted its analysis - this is further elaborated on in the receipt design section.

3.2 Edible Representation

In exploring edible representations, we began exploring drinks due to a thematic connection to journaling (e.g. writing at a cafe). We settled on hot chocolate mix due to its ease of use (powders were easier to deal with than liquids) and broad accessibility (it could be made vegan, gluten, and mostly allergen free).

We wanted to center flavor combinations that are typically unfamiliar in widely-available hot cocoa flavor mixes to augment user inquiry and attention to taste. For example, if their reflections mapped to stressful text, the user would get freeze-dried strawberries in their hot cocoa mix, which has bright, sharp, and fruity sweet qualities meant to delight and distract. While some flavors play particular roles in a dish (e.g. cinnamon is used to make dishes feel warm, sweet, and spiced), flavors do not necessitate particular emotions invoked since taste is often tied to memory and its associations are highly contextual from person to person. Higher "scores" on each measure corresponded to more servings.

These "flavors" were dispensed via servo motors and 3d printed dispensing machines displayed in 4. The flavors were crushed into powder/small chunks. They were further cut with milk powder to serve as a backdrop for the flavors and expand the volume of the powders so that the correct volume can be dispensed. The mapping of ingredients to emotions, along with their flavor characteristics, is presented in the table below.

Emotion	Flavor/ingredient	Characteristics
Rest	Toasted marshmallow	smooth, light, toasty
Stress	Freeze dried strawberry	intense, bright, fruity
Connection	Cinnamon	spicy, warm, familiar
Loneliness	Mint	icy, cold, slightly bitter

3.3 Receipt design

After receiving the drink mix, the user also received a receipt with a date, some explanation, flavor text, and an excerpt from a poem or story. For instance:

12/14/21

Our AI has read through every word that has ever existed and assessed every atom in your brain to provide an objective interpretation of your

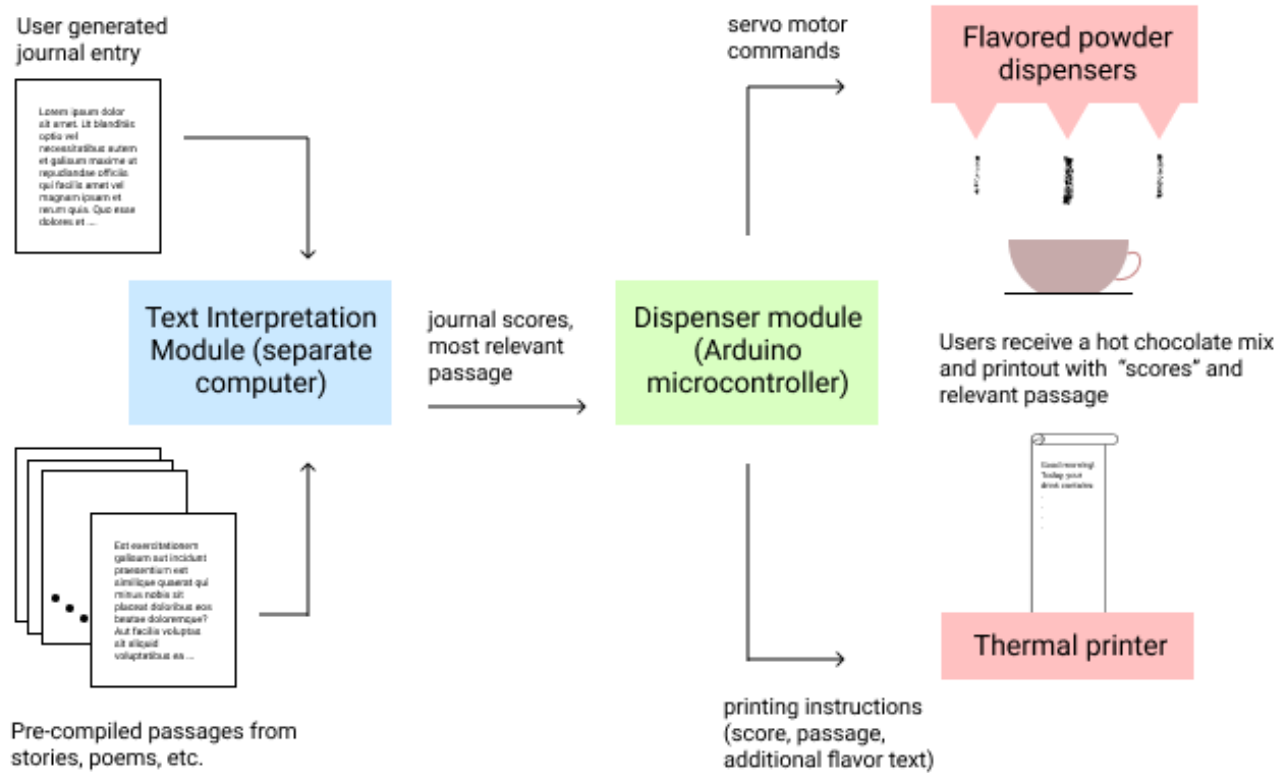


Figure 2: A system diagram of the Kukou Box inputs, outputs, and modules.

journal entry.

Today, your drink has:

3 connection (cinnamon)
-1 rest (strawberry)
2 things to chew on (chocolate chips)

While drinking, give this a read. It reminded us of your week.

"I came down with a horrible stomach bug when my husband was out of town. My best friend showed up with saltines, Sprite, essential oils, and the best part her Netflix password." - Meghan Simecek, Texas, to the Reader's Digest

We created this output for three reasons.

First, we found in our research that written communication was a powerful component in influencing the interpretation of flavor. In an ethnographic interview with Dandelion Chocolates, an artisan chocolatier in San Francisco that provides chocolate tasting experiences, we learned that peoples' interpretations of flavor were incredibly malleable to the text presented with the flavors. The text presented could affect peoples' recognition of the flavor notes

themselves as well as the emotional meaning associated with them. The text would help us build mappings of flavors to emotions.

Second, we wanted users to have some sense of how the computational output was generated. Rather than feel as if a machine had coldly and precisely evaluated their emotions, we wanted to playfully acknowledge the imprecision of computational interpretations of meaning and invite users to create their own. The tongue-in-cheek flavor text helped poke fun at the impossibility of "objective" evaluations. Furthermore, by presenting the most closely matching poem/story at the bottom of the receipt, users could have some sense for how the system "thought" about their journal entry beyond the assignment of numbers. This would further invite users to reflect lightly on the similarities between others' experiences and their own - or find meaning in their contrast.

Finally, the receipt was like a collectible in and of itself. In the group setting of the evaluation, people asked to see each others' receipts to understand each others' outputs (more details in preliminary findings); in an individual, long-use setting, someone could conceivably collect these receipts as objects to reminisce on in and of themselves.

4 PRELIMINARY FINDINGS

Our interaction design was piloted at our design department's Winter showcase with a total of 15 participants.



Figure 3: 3d printed dispensing mechanism with servo motor on the left; mounted dispensers on the inside of the box on the right.

4.1 Journaling

Participants were very eager to start journaling on the stationed laptop, knowing it would result in a cup of hot cocoa. We imagined people would feel uncomfortable writing personal entries in a public space, but found that people were indifferent. Some participants playfully acted out hiding gestures to pretend to keep the screen away from passerbyers peering, however, later opened up to other participants about their journal entries. Many participants chose to write about the stresses of their week or day, and even specific occasions and highlights such as meeting out of town friends and working through obstacles of a project.



Figure 4: A person utilizing our system during a user evaluation.

4.2 "How much connection did you get?" - Social Interaction

Most participants were accompanied by their friends and classmates who took turns and waited for each other as they journaled for hot cocoa. Once they all had their cup, we noticed a shift in how friend groups started to converse. Emulating the common conversations of daily updates and "how have you been?"s, we noticed that people started referring to emotions as drink flavors asking their friends "did you get connection in your hot cocoa?" in place of "how was your day?" Participants began equating hot cocoa flavors to emotional and daily adjectives. There were instances where friends predicted what flavors would represent their friend's week or day. One participant knew how stressed their friend was and suggested, "your drink will probably have a lot of strawberries," as strawberry was a flavor we mapped to remedy stress. Among another pair, as they tried each other's hot cocoas, one friend pointed out "guess you weren't stressed enough." Some participants would pay attention to the thickness of the drinks, comparing glasses and trying to make sense of how this was determined.

4.3 Individual Interactions

Depending on the components of the hot cocoa, some flavor to journal mapping entries were more obvious than others. The most commonly pointed out mapping being sweetness to positive entries. Participants who got sweeter and more cinnamon flavored hot cocoas reported being able to taste the entry more easily, referencing specific parts of their entry that might have resulted in sweetening their hot cocoa. One participant explained that her drink must have reflected her writing about getting lunch with her mom. Another participant discussed their feelings of excitement about the end of the semester and their project progress. As participants were given a printed receipt with their journal entry translated into quantities of emotions, some were surprised by the certain flavors and the amounts of it. This prompted participants to reflect and further discuss their entries as they sipped their hot cocoa. An interesting

experience was from a participant who received a “neutral” hot cocoa, where no flavors were added to a cup of chocolate chips. The participant wished he would have more cinnamon in his hot cocoa but understood it was not what he experienced the previous week, we discussed his plans for the next day and he said “definitely will be more cinnamon-y.” Four other participants described the bitterness intermixing with the sweet as surprisingly accurate, because “some parts of the week were hard, but other things were still sweet.”

4.4 Drinkable Reflection

Overall, our study has helped us realize the potential in tangible reflections, especially involving taste. Although flavors were not always apparent in what they represented in our Kukou box installation, when given a menu with flavors mapped to emotions, this guided conversations and opened space for reflection and conversation. Flavors were used to convey emotions once a key was provided. Hot cocoa, or taste in general, creates a new form of social interaction and self reflection.

5 CONCLUSION AND FUTURE WORK

In this paper, we introduced and evaluated KuKou Box, a novel and critical way in approaching reflection through personal informatics that utilized novel machine learning and edibilization techniques to transform journal-text to a drink and encourage users to reflect on their writings. We were successful in building a preliminary version of this system.

One future direction has to do with expanding the edibilization output itself to use more tastes and thus construct more taste-experience mappings. Taste is such a concrete sensation; we could synthesize new flavor profiles and tastes that could shed light on modes of reflection we never knew existed. For instance, what if people learned to map earthy spices, like turmeric, to feelings of confidence and stability? Might people say that a movie felt “earthy” the same way they might say it is “bittersweet”?

Another future direction we aim to take involves more long term studies with people. As we’ve found that providing a conversation structure using flavors can give people an opportunity to express their thoughts and emotions in a different medium, this is necessary to see if peoples’ reflective modes are themselves changed through use of this system, and to understand how edibilization affects one’s experiences in the long term.

REFERENCES

- [1] Nazanin Andalibi and Justin Buss. 2020. *The Human in Emotion Recognition on Social Media: Attitudes, Outcomes, Risks*. Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3313831.3376680>
- [2] Eric P.S. Baumer, Vera Khovanskaya, Mark Matthews, Lindsay Reynolds, Victoria Schwanda Sosik, and Geri Gay. 2014. Reviewing Reflection: On the Use of Reflection in Interactive System Design. In *Proceedings of the 2014 Conference on Designing Interactive Systems* (Vancouver, BC, Canada) (DIS '14). Association for Computing Machinery, New York, NY, USA, 93–102. <https://doi.org/10.1145/2598510.2598598>
- [3] Abeba Birhane. 2021. The Impossibility of Automating Ambiguity. *Artificial Life* 27, 1 (2021), 44–61.
- [4] Rafael A. Calvo and Dorian Peters. 2013. The Irony and Re-Interpretation of Our Quantified Self. In *Proceedings of the 25th Australian Computer-Human Interaction Conference: Augmentation, Application, Innovation, Collaboration* (Adelaide, Australia) (OzCHI '13). Association for Computing Machinery, New York, NY, USA, 367–370. <https://doi.org/10.1145/2541016.2541070>
- [5] Markéta Dolejšová and Tereza Lišková. 2015. StreetSauce: Taste Interaction and Empathy with Homeless People. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems* (Seoul, Republic of Korea) (CHI EA '15). Association for Computing Machinery, New York, NY, USA, 1247–1252. <https://doi.org/10.1145/2702613.2732777>
- [6] Florian 'Floyd' Mueller, Tim Dwyer, Sarah Goodwin, Kim Marriott, Jialin Deng, Han D. Phan, Jionghao Lin, Kun-Ting Chen, Yan Wang, and Rohit Ashok Khot. 2021. *Data as Delight: Eating Data*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3411764.3445218>
- [7] William W. Gaver, Jacob Beaver, and Steve Benford. 2003. Ambiguity as a Resource for Design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Ft. Lauderdale, Florida, USA) (CHI '03). Association for Computing Machinery, New York, NY, USA, 233–240. <https://doi.org/10.1145/642611.642653>
- [8] Tom Gayler, Corina Sas, and Vaiva Kalnikaite. 2020. Co-Designing Flavor-Based Memory Cues with Older Adults. In *Companion Publication of the 2020 International Conference on Multimodal Interaction* (Virtual Event, Netherlands) (ICMI '20 Companion). Association for Computing Machinery, New York, NY, USA, 287–291. <https://doi.org/10.1145/3395035.3425644>
- [9] Gabriel Grill and Nazanin Andalibi. 2021. Attitudes and Folk Theories of Data Subjects on Transparency and Accuracy in Emotion Recognition. In *Proceedings of The 25th ACM Conference on Computer-Supported Cooperative Work and Social Computing*. Association for Computing Machinery, New York, NY, USA. <https://grill.net/assets/pdf/emotion-recognition-accuracy-transparency-preprint.pdf>
- [10] Rohit Ashok Khot, Ryan Pennings, and Florian 'Floyd' Mueller. 2015. EdIPulse: Supporting Physical Activity with Chocolate Printed Messages. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems* (Seoul, Republic of Korea) (CHI EA '15). Association for Computing Machinery, New York, NY, USA, 1391–1396. <https://doi.org/10.1145/2702613.2732761>
- [11] Yubo Kou, Xinning Gui, Yunan Chen, and Bonnie Nardi. 2019. *Turn to the Self in Human-Computer Interaction: Care of the Self in Negotiating the Human-Technology Relationship*. Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3290605.3300711>
- [12] Ian Li, Anind K. Dey, and Jodi Forlizzi. 2011. Understanding My Data, Myself: Supporting Self-Reflection with Ubicomp Technologies. In *Proceedings of the 13th International Conference on Ubiquitous Computing* (Beijing, China) (UbiComp '11). Association for Computing Machinery, New York, NY, USA, 405–414. <https://doi.org/10.1145/2030112.2030166>
- [13] Ian Li, Jodi Forlizzi, and Anind Dey. 2010. Know Thyself: Monitoring and Reflecting on Facets of One's Life. In *CHI '10 Extended Abstracts on Human Factors in Computing Systems* (Atlanta, Georgia, USA) (CHI EA '10). Association for Computing Machinery, New York, NY, USA, 4489–4492. <https://doi.org/10.1145/1753846.1754181>
- [14] George Loewenstein and David Schkade. 1999. Wouldn't it be nice? Predicting future feelings. *Well-being: The foundations of hedonic psychology* (1999), 85–105.
- [15] Jos Mojet and Ep Köster. 2016. Flavor memory. In *Multisensory Flavor Perception*. Elsevier, 169–184.
- [16] Masashi Nakagawa, Keiko Mizuma, and Takako Inui. 1996. Changes in taste perception following mental or physical stress. *Chemical senses* 21, 2 (1996), 195–200.
- [17] H. Oku, M. Nomura, K. Shibahara, and A. Obara. 2018. Edible Projection Mapping. In *SIGGRAPH Asia 2018 Emerging Technologies* (Tokyo, Japan) (SA '18). Association for Computing Machinery, New York, NY, USA, Article 2, 2 pages. <https://doi.org/10.1145/3275476.3275486>
- [18] Gaurav Patekar, Karan Dudeja, Himanshu Bablani, and Debanshu Bhaumik. 2018. Data Jalebi Bot. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI EA '18). Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3170427.3186528>
- [19] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. arXiv:1908.10084 [cs.CL]
- [20] Kimiko Ryokai, Elena Durán López, Noura Howell, Jon Gillick, and David Baman. 2018. *Capturing, Representing, and Interacting with Laughter*. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3173574.3173932>
- [21] Yun Wang, Xiaojuan Ma, Qiong Luo, and Huamin Qu. 2016. Data Edibilization: Representing Data with Food. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (San Jose, California, USA) (CHI EA '16). Association for Computing Machinery, New York, NY, USA, 409–422. <https://doi.org/10.1145/2851581.2892570>
- [22] Philip Weber, Kevin Krings, Julia Nießner, Sabrina Brodesser, and Thomas Ludwig. 2021. *FoodChattAR: Exploring the Design Space of Edible Virtual Agents for Human-Food Interaction*. Association for Computing Machinery, New York, NY, USA, 638–650. <https://doi.org/10.1145/3461778.3461998>